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Nanoimprint lithography: challenges and prospects

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Abstract

We review the salient aspects of nanoimprint lithography and consider the challenges it faces in becoming a standard fabrication technique, such as costs and throughput. We discuss material issues such as visco-elasticity and functionality of the printed material. By way of an illustration, we present printing results of 50 nm features over a $2 \times 2 \text{ cm}^2$ area which are reproducible with high fidelity. Data of printing 15 nm features in PMMA using a Cr stamp was obtained.

1. Introduction

The application of nanofabrication to printing gives rise to a need to develop a competitive parallel process, which may serve the technological demands of nanoelectronics and related areas, while bearing in mind reliability, throughput and cost. In this field, the work of Chou *et al* [1] is widely considered as being pioneering among the nanofabrication community.

In what follows, the basic principles, bottlenecks and prospects of nanoimprint lithography will be discussed. One important aspect concerning a polymer's suitability for printing is understanding its visco-elasticity behaviour. In this paper we discuss various conformational changes of polymers from a printing point of view.

To test nanoimprinting at less than 50 nm we performed experimental printing of 50 nm features over a $2 \times 2 \text{ cm}^2$ square and show the preliminary results of printing at 15 nm in PMMA. These tests allow us to point out some issues that need further research.

2. The nanoimprinting process

A stamp with the desired nanometre-range features is needed. This is usually fabricated by, for example, optical lithography and/or electron-beam lithography and dry etching. The material to be printed, a printable polymer, is spun onto a solid substrate. The stamp and the substrate are placed on parallel stages (see figure 1). These are heated up to the printing temperature, which is above the glass temperature of the polymer to be moulded and a certain amount of time is

allowed for the thermal equilibrium to be reached. The stamp and substrate are brought into physical contact and pressure is applied, followed by subsequent cooling. The temperature and pressure holding times depend on several factors and are typically a few minutes. Demounting and separation of the stamp and substrate takes place when both are just below the glass temperature. The polymer thickness contrast left on the substrate after printing can be used as: (a) a mask for dry etching if the imprinted polymer is resistant enough, (b) a step in a lift off process, or (c) assuming the printed polymer has a functionality such as conductivity, optical linear or nonlinear response, among others, as a device itself. There are several key issues which have to be considered for imprint lithography to make it a competitive nanofabrication technology.

2.1. Multilevel capability

The main criticism levied at imprint lithography is the multilevel issue. For this to be overcome the first step is to develop an in-plane alignment approach. The target is alignment better than 10 nm, at least for VLSI applications. In general, the stamp is considered to be the functional equivalent of the photo-mask in conventional projection lithography. At present the limit is $\sim 1 \mu\text{m}$, as shown by the step-and-stamp process, using a commercially available stepper [2]. In a first approximation alignment depends on the stamp size, the thermal and mechanical stability of the polymer during alignment and the choice of stamp and substrate material. Optical alignment techniques are also limited to about $1 \mu\text{m}$. Piezo-driven techniques allow better resolution (well below

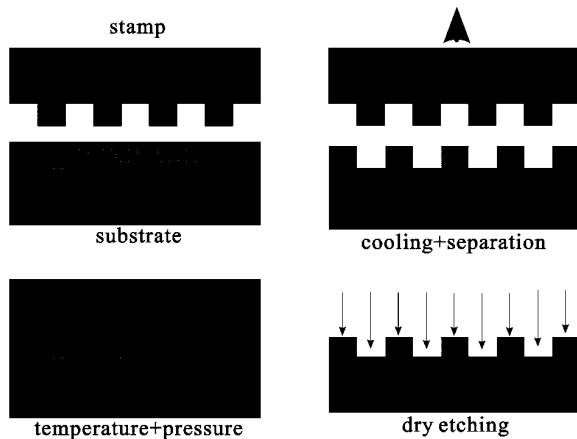


Figure 1. Schematic of the nanoimprinting process.

100 nm) but only for stamps of a few millimetres. It is estimated that, by considering thermal effects, the bending and specifications of the equipment, piezo-driven approaches could provide an alignment of about 30 nm over 50 mm.

Using a printing machine developed in-house aided by a commercially available aligner and matching the thermal expansion coefficients of the stamp and substrate, alignment to within just under $1\text{ }\mu\text{m}$ over a 4" wafer has been demonstrated which printed gratings with 100 nm features [3]. A proposal has been recently put forward [4] for an alignment system suitable for printing with UV-curing, which combines an adaptive wafer holder with the optical detection of diffraction fringes.

2.2. Throughput

Whereas a working definition of throughput as understood in conventional lithography is still lacking, a rough approximation would consider the actual printing and alignment times, which are both currently in the range of a few minutes. The time required for stamp cleaning and replacement and, if needed, coating with an antisticking layer after a number of prints will then all have to be included, although at present these values differ substantially as most results are obtained using non-automated processes. Factors leading to an increase of throughput include:

- (a) *Large stamp size.* The stamp size determines the area to be printed each time. Ideally this will have a size compatible with standard production equipment. Some potential drawbacks include the parallelity of the wafer giving rise to thermal gradients in printing, which can in principle be addressed with a suitably designed printing machine. Recently, 6" wafers have been printed in a compact disc-like production setting. The 6" stamp had representative areas with feature sizes down to 50 nm, separated by several hundreds of nanometres from the next feature [5]. One technique which addresses the large-area stamp issue and other factors is step and flash imprint lithography [6], which has been tested with stamps up to 2" and a test-bed machine for 8" wafers has been developed [6].
- (b) *Stamps with high feature density.* Depending on the application used, one of the limits is the flow of the displaced polymer. To date, precise design rules

concerning negative and positive features in a given stamp are still under development. In the step-and-stamp variation of printing, 150 mm^2 were printed with features of 400 nm separated by 400 nm spaces in 36 consecutive stamping steps [2]. Imprint tests on areas of $200 \times 200\text{ }\mu\text{m}^2$ containing 50 nm features separated by 50 nm spaces are discussed in section 4.

- (c) *Sticking.* Ideally, the use of an antisticking layer in the printing process is to be avoided. Much research is in progress to investigate polymers and define processing windows, which make the use of the antisticking layer redundant. Here the choice of printing temperature, the visco-elastic properties of the polymer and the interfacial energy between the polymer and substrate and between the polymer and stamp are among the key factors [7].
- (d) *Curing.* If curing the polymer is needed then a reduction in the curing time is important. The aim here is to move away from a curing step, either by temperature or UV illumination, unless (a) it takes place in combination with another nanofabrication technique, which would then compensates for the extra time, or (b) it adds flexibility to the process [8]. Such flexibility could be the result of the use of bi- or tri-layers of polymers with different printing and curing behaviour.
- (e) *Printing temperature and pressure.* Both parameters should be as low as possible in view of the time needed for temperature and pressure cycling. Current research in polymers for printing has yielded two new polymers, which are commercially available from mrt GmbH. These are mr-I-8000 and mr-I-9000, which have glass temperatures of 107 and 63°C , respectively (other specifications available from the suppliers). At present knowledge about these polymers is insufficient and cannot supply a fixed window of parameters, since the medium-term mechanical recovery (see section 3) of sub-20 nm features has not been systematically studied. Pressure is less important, since its application only takes a minute or so. More significant may be the rate at which pressure is increased, again with respect to the mechanical recovery of the polymer.
- (f) *Stamp lifetime.* This issue has not been explored to its limits because most tests take place in a laboratory environment and are non-automated. Manually controlled experiments have been reported of trials repeated up to 36 times using a flip-chip bonder [9]. One aspect which merits attention is the possibility of using printing to fabricate or, more appropriate, to replicate stamps. This approach has recently been investigated [10] showing that by using a thermoset polymer in its pre-polymer state, subsequently followed by thermal cross-linking, stamps with 400 nm features can be successfully replicated.

2.3. Standards

Some other aspects affecting all technologies known generically as 'printing' are validation and standards. Validation, or the quality control issue, needs agreement as to what counts as tolerances for a good print. Optical diffraction techniques lend themselves to monitoring the quality of periodic structures, but there is a need for fast quality control

methods in printing. Standards depend strongly on design rules and it is too early to expect definitive statements.

3. Polymer issues

Good control of the nanoimprint process requires some understanding of the underlying mechanical properties of polymers. The combination of viscous and elastic behaviour in a polymer is not a simple superposition of both behaviours, but brings about a new phenomenon named anelasticity [11]. It is apparent from observation that part of the deformation, although being reversible, requires some time to establish itself. To illustrate the different contributions to the deformation we discuss the response of a polymer to an external force.

The immediate deformation of the polymer corresponds to its elastic response. The elastic contribution to the deformation is fully recovered upon unloading of the polymer. A second contribution corresponds to the viscous response of the system to external stress. The viscous behaviour is irreversible and its contribution to the overall deformation can be determined from the recovery curve. Finally, there is a retarded deformation, which requires some time to establish itself and is fully recovered some time after unloading the polymer. The occurrence of a retarded deformation shows that visco-elasticity is more than a simple superposition of viscous and elastic contributions. The retarded deformations are frequently addressed as anelastic behaviour and are related to the flexibility of the macromolecules. Polymers combine viscous, elastic and anelastic properties and this behaviour is generally addressed as 'visco-elasticity'. The relative weights of the three contributions to the overall deformation depend on the type of polymer and particularly on the temperature. It is clear that an understanding of the latter is essential for good control of the imprint process since only irreversible deformations are desirable in nanoimprinting.

With respect to the mechanical behaviour of amorphous polymers, there are two important points to be considered: (i) the conformational changes of the chain that allow for both local motions of short chain segments and motions of entire chains understood as thermally activated processes and therefore taking place on typical timescales which depend strongly on temperature, and (ii) the formation of a temporary network of large, entangled chains. We will discuss the temperature dependence of the mechanical behaviour of polymers, but the response of a polymer to external forces depends strongly on the temperature-dependent timescales of the different motions.

At low temperatures the conformational changes are very sluggish, taking place on timescales of hours to years. The mayor contribution to the response comes from the elongation of the atomic distances within and between the macromolecules and the response to external forces corresponds to that of an ideal elastic body. The amorphous polymer shows the behaviour of a truly glassy, hard and brittle material. This region is referred to as the *glassy region*.

At temperatures near the glass temperature short range motions of chain segments due to conformational changes will take place over accessible timescales. On increasing the temperature the energy barrier for conformational changes

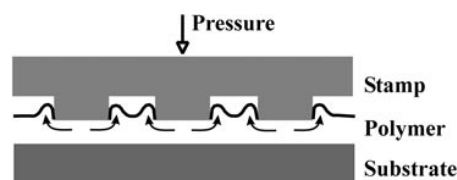


Figure 2. Schematic depicting polymer flow during nanoimprinting.

related to such motions will be crossed more frequently and local motions become faster. However, the temperature is not sufficiently high enough to permit the motion of entire chains within the experimental timescales involved. Hence, a temporary network of entangled chains has to be considered. Under applied external forces the chain segment fixed between two entanglement points will be extended due to local conformational changes. To establish this extension some time is required and the overall deformation will exhibit a retarded contribution. This region is called the *transition region*.

The region that follows the transition region as the temperature increases further, is the *rubber-elastic region*. Now the temperature is sufficiently high, and thus, local motions of chain segments take place on very short timescales, whereas entire chains are still fixed by the temporary entanglement network. On application of external forces to the polymer, the changes in conformation are fast enough to allow for the extension of chain segments fixed between entanglement points and the corresponding large deformations are observed.

Finally, the *region of terminal flow* is reached on increasing the temperature. Now, even the motion of entire chains takes place on acceptable timescales and the polymer flows by chain sliding. This region is of particular interest for nanoimprinting because this behaviour within the experimental timescales is mainly described by irreversible flow, which is the desired behaviour for pattern transfer (figure 2). A polymer melting at sufficiently high temperature will basically show a viscous flow, except when the external forces change in timescales shorter than the time required by the chains to slide across the entanglement points.

The following consideration has some important consequences for nanoimprinting: even in the terminal flow region the action of restoring forces which tend to recover the initial shape of the polymer has to be taken into account. These forces will act on the mould and may cause sticking-like behaviour during the separation of the stamp from the polymer [12].

Another important point is the dependence on molecular weight (MW). Since the glass transition depends strongly on local motion due to conformational changes, it is expected that the transition temperature does not depend strongly on the MW. On the other hand, the number of entanglements per chain depends on the MW of the sample and thus it might be expected that at a given temperature a polymer of low MW will flow easier than one with a high MW. For a high MW an extended rubber-elastic plateau is expected [12].

4. Nanoimprint testing of small features

In this section the capabilities of nanoimprinting to fabricate small features on PMMA are illustrated using stamps fabricated by electron-beam lithography.

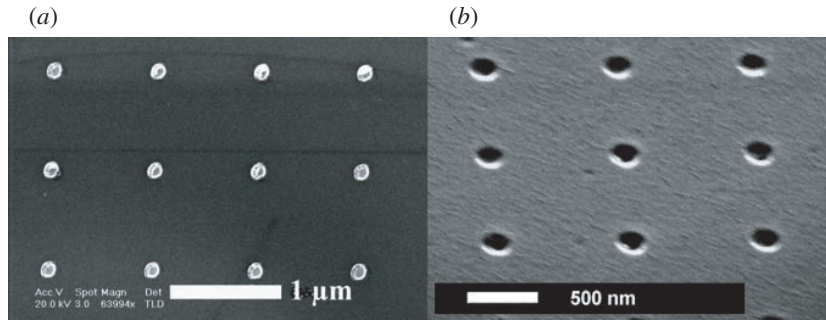


Figure 3. Scanning electron microscopy (SEM) micrographs of a $200 \times 200 \mu\text{m}^2$ Cr stamp with 60 nm high columns 120 nm in diameter with a $1 \mu\text{m}$ period (a) and an imprint into a 100 nm-thick PMMA layer (b). After imprinting a 30 nm gold layer was deposited for SEM imaging.

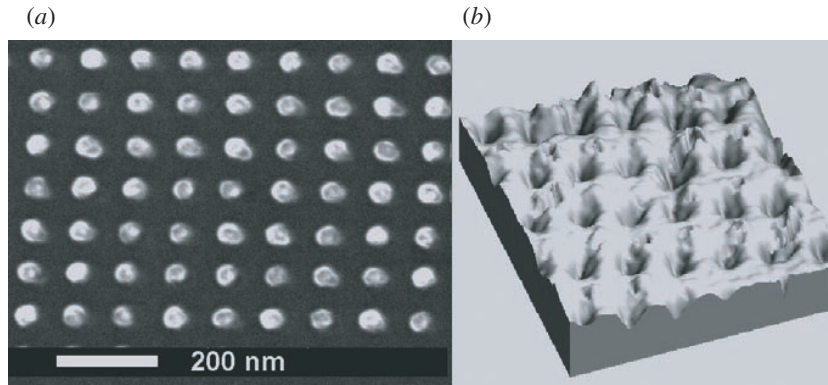


Figure 4. SEM micrographs of a Cr stamp consisting of 60 nm high columns 50 nm in diameter with a 100 nm period (a) and an AFM image of 50 nm features imprinted into a 100 nm-thick PMMA layer (b). The stamp has several areas of $200 \times 200 \mu\text{m}^2$ repeated over $2 \times 2 \text{ cm}^2$.

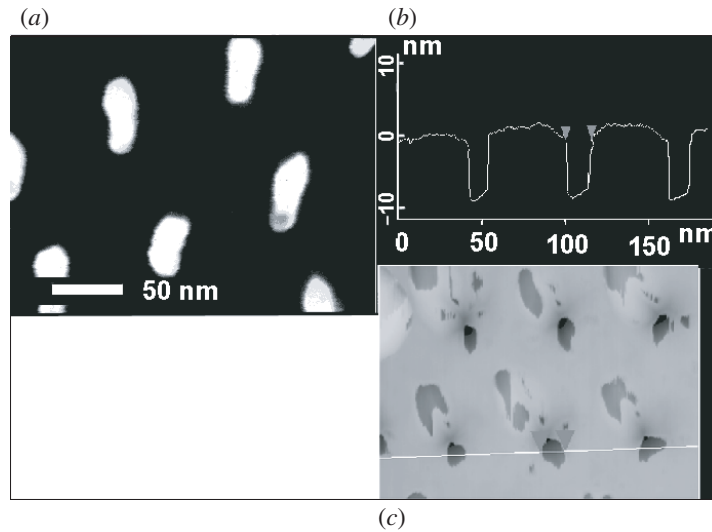


Figure 5. SEM micrograph of a Cr stamp with 15 nm feature size and 60 nm high (a). AFM image (c) section analyses (b) of the corresponding imprint. PMMA with a MW of 200 000 was used.

4.1. Stamp fabrication

Stamps containing features less than 120 nm in size were fabricated on a silicon substrate using electron-beam lithography, Cr evaporation and lift-off. Two kinds of stamps with different features density were designed, in particular with

1 : 8 and 1 : 2 ratios of feature size to the period.

The first stamp consists of 70 nm high columns 120 nm in diameter with a $1 \mu\text{m}$ period. A SEM micrograph of this stamp is shown in figure 3(a). The electron-beam resist used was a double-layer PMMA film with MW of 200 000 (bottom

layer) and 950 000 (top layer) with thickness of 160 and 60 nm, respectively. After spinning each layer the substrate was baked at 170°C for 30 min.

The stamp features were written on $200 \times 200 \mu\text{m}^2$ squares using a Philips XL 30S FEG SEM equipped with a RAITH lithography module. For exposure the following conditions were chosen: accelerating voltage, 20 kV; beam current, 26.6 nA; and resist sensitivity, $480 \mu\text{As cm}^{-2}$. After exposure the resist was developed over 30 s in a standard 4-methyl-2-pentanone : 2-propanol (1:3) developer. A 60 nm-thick Cr layer was deposited at a rate of 0.04 nm s^{-1} using an electron-beam evaporator. Lift-off was performed in the AR 600-70 remover under ultrasonic agitation.

The second stamp consists of 60 nm high columns 50 nm in diameter with a 100 nm period. A SEM micrograph of the stamp shown in figure 4(a). There are several differences in the fabrication process, which are connected with a stronger proximity effect. A one-layer 100 nm thick PMMA film with MW of 950 000 was used. The written structure is a pattern of single pixel dots, each exposed for 0.01 ms.

4.2. Imprinting results

The imprints were carried out using a small hydraulic press cooled by oil in which samples can be heated up to 240°C and the pressure can be stabilised between 20 and 200 bar. The time required for one cycle (heating, cooling, separation) is between 10 and 20 min, depending on the imprint temperature and the temperature at which the stamp is separated from the substrate. The size of stamps was limited to $2 \times 2 \text{ cm}^2$ squares in our equipment.

The imprints were made into poly(methyl methacrylate). A 100 nm PMMA film was spun onto a RCA cleaned silicon substrate and annealed for 30 min at 170°C. The imprint temperature was 190°C, i.e. 95°C above the glass temperature.

Features 120 nm in lateral size with a 1 μm period were imprinted into PMMA with MW of 950 000. A SEM micrograph of a typical imprint is shown in figure 3(b). A good pattern transfer can be seen, filling completely the cavities and showing no sticking over the whole area. However, this is not always the case when imprinting 50 nm features with 50 nm spaces. The imprints sometimes show insufficient polymer flow and some sticking effects. It was found that the MW of the PMMA also plays a role and thus imprints were made using PMMA with a MW of 200 000. An AFM image of a successful imprint of such features is shown in figure 4(b). This result is in agreement with the expected dependence of polymer flow upon MW as discussed in section 3.

For a further reduction in feature size to below that of the polymer molecule, it is necessary to understand the role of polymer-chain deformation and polymer flow. For this purpose AFM investigations into large imprinted surface with a resolution of less than 1 nm are needed.

Recently, a $3 \times 3 \text{ mm}^2$ Cr stamp was fabricated, containing elliptical features with a lateral size of around 15 nm in one direction and 60 nm high (see figure 5(a)). The imprint was

performed onto a 200 000 PMMA film 100 nm thick. An AFM image and line scan are shown in figure 5(b).

5. Conclusions

The throughput issue of nanoimprinting sets look to be resolved since: (a) the size of successfully printed samples is 6", as shown by Lund University; (b) the resolution achieved is 50 nm; and (c) the printing time is about 20 min (including alignment, heating and cooling cycles). Using these values the 'exposure rate', which is a measure of throughput, for 50 nm features reaches a value of $0.152 \text{ cm}^2 \text{ s}^{-1}$, making it comparable to scattering with angular limitation projection electron-beam lithography (SCALPEL). Moreover, the use of an antisticking layer can be avoided by choosing suitable printing parameters and polymers.

The choice of polymers appropriate for nanoimprinting can be aided by considering the mechanical behaviour discussed here. Good flow behaviour and antisticking properties are essential for nanoimprinting and are characteristics exhibited by commercially available polymers.

The possibility of nanoimprinting 15 nm features with a stamp of $3 \times 3 \text{ mm}^2$ was demonstrated. To obtain a good pattern transfer polymers with low MW are recommended.

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